DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

YEAR III SEMESTER II **SUPPLEMENTARYRDINARY** EXAMINATION FOR THE BACHEROR OF BUSINESS INFROMATION TECHNOLOGY

ICS 2311: COMPUTER GRAPHICS

DATE: January 2016 **TIME: 2 HOURS**

Instructions: Answer Question 1 and Any Other Two.

Question 1: (30 Marks)

a) Using sample OpenGL statements explain the general syntax for rendering both a solid and a wire sphere centered at the modeling coordinates origin of a specified radius and subdivided around the Z axis into slices and along the Z axis into stacks. **(4mks)**

b) Explain the functions of the following components in the operation of raster displays. **(4mks)**
   i). Frame buffer.
   ii). Video controller

c) Write a program that uses a call back function to render a grey window on the screen. Include functions to indicate the window size and position on the screen. **(5mks)**

d) Graphic illustrations are used in presentations to help convey an idea or express a mood. Distinguish between **vector** and **Bitmap** illustrations graphics giving one advantage and one disadvantage of each. **(4mks)**

e) Animation adds to graphics the dimension of time which vastly increases the amount of information to be transmitted.

   i) Historical approach of animation has classified computer animation into two main categories. Briefly describe the two categories. **(4mks)**

.

   ii) Using well labelled diagrams differentiate between single buffering and double buffering in animated graphics using OpenGL renderer. **(6mks)**

   iii) Briefly describe the purpose of the function **glutSwapBuffers()** in rendering animated graphics in openGL. **(2mks)**

f) What is scan conversion? **(1mk)**

Question 2: (20 marks)

a) Outline the basic steps involved in converting Cartesian coordinate system to the system understandable points. **(6mks)**

b) Differentiate between random scan and raster scan monitors . **(4mks)**

c) Explain the purpose of each of the following OpenGl geometric primitives. **(4mks)**

i) GL_LINE_STRIP
ii) GL_TRIANGLE_STRIP
iii) GL_QUAD_STRIP
iv) GL_POLYGON

d) Study the OpenGL code snippet below and sketch the graphic that will render when executed. **(6mks)**

```
glBegin(GL_TRIANGLE_STRIP);
glColor3f(1,1,1); // color
glVertex2f(0,0); //v1
glVertex2f(0,1); // v2
glVertex2f(1,0); // v3
glVertex2f(1,1); // v4
glVertex2f(2,0); // v5
glEnd();
```

Question 3: (20 marks)

a) Differentiate between Quadtree and Octree data structures as used in computer graphics. **(4mks).**

b) Explain the function of each of the following OpenGl functions as used for drawing primitives **(4marks)**

i) glBegin(mode);
ii) glColor3f(1,0,0);
iii) glVertex3f(0,1.5,-2);
iv) glEnd(void);

c) An openGL callback function is different from a normal function as it is never called directly by the application.

i) Briefly explain the purpose of the ***glutDisplayFunc()*** call back function. **(2mks)**

ii) Write a program to demonstrate how the ***glutDisplayFunc()*** call back function is used in openGL programs. **(6mks)**

d) Using a well labelled diagram, explain how a refresh CRT display works. **(4mks)**

Question 4: (20 marks)

a) Using the relevant OpenGL syntax for their default values, demonstrate the difference between the following lighting model components and provide the general syntax for setting their RGB default values. **(6mks)**
i) Ambient
ii) Diffuse
iii) Specular

b) Explain the following terms as used in 3D graphics models. **(6marks)**
i) Rendering

ii) Ray tracing
iii) Pixel-value

c) Describe the following 3D display techniques. **(6mks)**

i) Perspective projection

ii) Parallel projection

iii) Cueing

d) Even though the rasterization tries to generate a completely straight line, yet in few cases we may not get equal brightness. Explain any two compromises that are made while generating such lines. **(2mks)**

Question 5: (20 marks)

a) Consider two raster systems with resolutions 1024 by 780 and 1340 by 1010.

i) What is the standard Aspect Ratio in use by the two raster systems described above? **(4marks)**

ii) Determine the number of pixels that should be accessed per second if the display controller refreshes the screen at a rate of 30Hz. **(2mks)**

iii) Determine the number of pixels that should be accessed per scan cycle if the system uses the interlacing technique. **(2mks)**

iv) Determine the minimum difference in storage capacity (per second) between the two systems if each system is defined using 24 bits. Give your solution in kilobytes.**4mks)**

b) i) Outline one advantage and one disadvantage of the DVST monitors. **(2mks)**

ii) Giving an example in each case, discuss any three types of models that can be implemented using computer graphics. **(6mks)**